

Characterization of extended-spectrum cephalosporin-resistant *Salmonella enterica* serovar Heidelberg isolated from food animals, retail meat, and h...



Salmonella enterica is one of the most common causes of foodborne illness in the United States. Although salmonellosis is usually self-limiting, severe infections typically require antimicrobial treatment, and ceftriaxone, an extended-spectrum cephalosporin (ESC), is commonly used in both adults and children. Surveillance conducted by the National Antimicrobial Resistance Monitoring System (NARMS) has shown a recent increase in ESC resistance among *Salmonella* Heidelberg isolated from food animals at slaughter, retail meat, and humans. ESC resistance among *Salmonella* in the United States is usually mediated by a plasmid-encoded bla(CMY) β -lactamase. In 2009, we identified 47 ESC-resistant bla(CMY)-positive Heidelberg isolates from humans (n=18), food animals at slaughter (n=16), and retail meats (n=13) associated with a spike in the prevalence of this serovar. Almost 90% (26/29) of the animal and meat isolates were isolated from chicken carcasses or retail chicken meat. We screened NARMS isolates for the presence of bla(CMY), determined whether the gene was plasmid-encoded, examined pulsed-field gel electrophoresis patterns to assess the genetic diversities of the isolates, and categorized the bla(CMY) plasmids by plasmid incompatibility groups and plasmid multi-locus sequence typing (pMLST). All 47 bla(CMY) genes were found to be plasmid encoded. Incompatibility/replicon typing demonstrated that 41 were IncI1 plasmids, 40 of which only conferred bla(CMY)-associated resistance. Six were IncA/C plasmids that carried additional resistance genes. pMLST of the IncI1-bla(CMY) plasmids showed that 27 (65.8%) were sequence type (ST) 12, the most common ST among bla(CMY)-IncI1 plasmids from Heidelberg isolated from humans. Ten plasmids had a new ST profile, ST66, a type very similar to ST12. This work showed that the 2009 increase in ESC resistance among *Salmonella* Heidelberg was caused mainly by the dissemination of bla(CMY) on IncI1 and IncA/C plasmids in a variety of genetic backgrounds, and is likely not the result of clonal expansion.